

# Transmission Line Characteristics

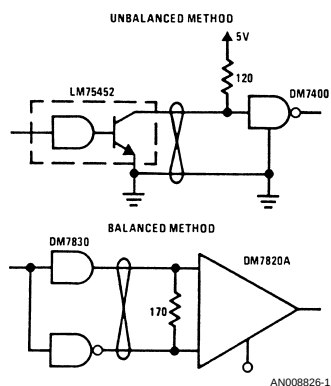
National Semiconductor  
Application Note 108  
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## INTRODUCTION

Digital systems generally require the transmission of digital signals to and from other elements of the system. The component wavelengths of the digital signals will usually be shorter than the electrical length of the cable used to connect the subsystems together and, therefore, the cables should be treated as a transmission line. In addition, the digital signal is usually exposed to hostile electrical noise sources which will require more noise immunity than required in the individual subsystems environment.

The requirements for transmission line techniques and noise immunity are recognized by the designers of subsystems and systems, but the solutions used vary considerably. Two widely used example methods of the solution are shown in *Figure 1*. The two methods illustrated use unbalanced and balanced circuit techniques. This application note will delineate the characteristics of digital signals in transmission lines and characteristics of the line that effect the quality, and will compare the unbalanced and balanced circuits performance in digital systems.



## NOISE

The cables used to transmit digital signals external to a subsystem and in route between the subsystem, are exposed to external electromagnetic noise caused by switching transients from actuating devices of neighboring control systems. Also external to a specific subsystem, another subsystem may have a ground problem which will induce noise on the system, as indicated in *Figure 2*.

The signals in adjacent wires inside a cable may induce electromagnetic noise on other wires in the cable. The induced electromagnetic noise is worse when a line termi-

nated at one end of the cable is near to a driver at the same end, as shown in *Figure 3*. Some noise may be induced from relay circuits which have very large transient voltage swings compared to the digital signals in the same cable. Another source of induced noise is current in the common ground wire or wires in the cable.

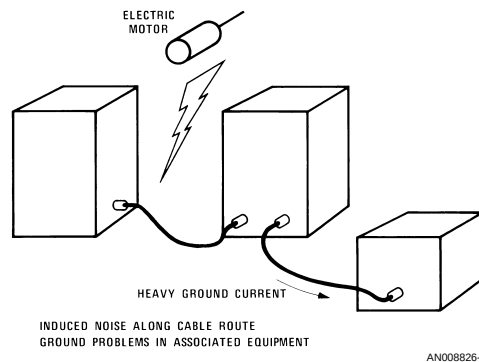


FIGURE 2. External Noise Sources

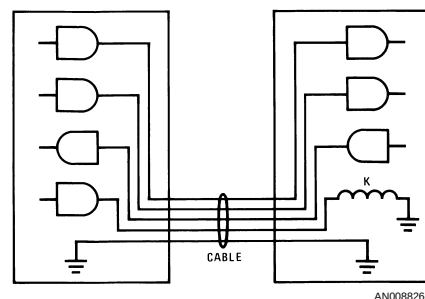


FIGURE 3. Internal Noise Sources

## DISTORTION

The objective is the transmission and recovery of digital intelligence between subsystems, and to this end, the characteristics of the data recovered must resemble the data transmitted. In *Figure 4* there is a difference in the pulse width of the data and the timing signal transmitted, and the corresponding signal received. In addition there is a further differ-

ence in the signal when the data is "AND"ed with the timing signal. The distortion of the signal occurred in the transmission line and in the line driver and receiver.

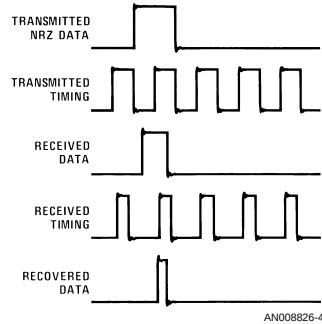


FIGURE 4. Effect of Distortion

A primary cause of distortion is the effect the transmission line has on the rise time of the transmitted data. Figure 5 shows what happens to a voltage step from the driver as it travels down the line. The rise time of the signal increases as the signal travels down the line. This effect will tend to affect the timing of the recovered signal.

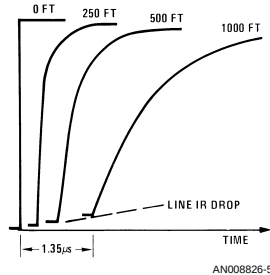


FIGURE 5. Signal Response at Receiver

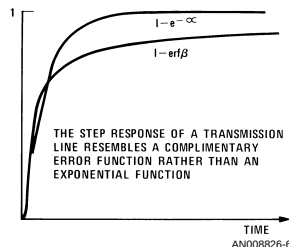


FIGURE 6. Signal Rise Time

The rise time in a transmission line is not an exponential function but a complementary error function. The high frequency components of the step input are attenuated and delayed more than the low frequency components. This attenuation is inversely proportional to the frequency. Notice in

Figure 6 particularly that the signal takes much longer to reach its final DC value. This effect is more significant for fast risetimes.

The Duty Cycle of the transmitted signal also causes distortion. The effect is related to the signal rise time as shown in Figure 7. The signal doesn't reach one logic level before the signal changes to another level. If the signal has a  $\frac{1}{2}$  (50%) Duty Cycle and the threshold of the receiver is halfway between the logic levels, the distortion is small. But if the Duty Cycle is  $\frac{1}{8}$  as shown in the second case the signal is considerably distorted. In some cases, the signal may not reach the receiver threshold at all.

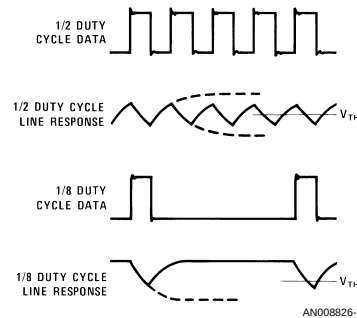


FIGURE 7. Signal Distortion Due to Duty Cycle

In the previous example, it was assumed that the threshold of the receiver was halfway between the ONE and ZERO logic levels. If the receiver threshold isn't halfway the receiver will contribute to the distortion of the recovered signal. As shown in Figure 8, the pulse time is lengthened or shortened, depending on the polarity of the signal at the receiver. This is due to the offset of the receiver threshold.

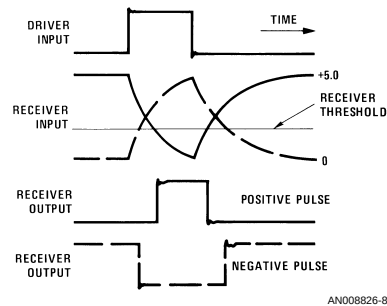
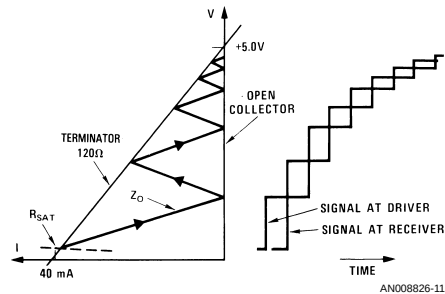


FIGURE 8. Slicing Level Distortion

#### UNBALANCED METHOD

Another source of distortion is caused by the IR losses in the wire. Figure 9 shows the IR losses that occur in a thousand feet of no. 22 AWG wire. Notice in this example that the losses reduce the signal below the threshold of the receiver in the unbalanced method. Also that part of the IR drop in the

time diagrams, there are transient voltage and current signals that subtract from the particular signal and add to the system noise.



### FIGURE 9. Unbalanced Method



**FIGURE 10. LM75451, DM7400 Line Voltage Waveforms**

## BALANCED METHOD

Figure 12 shows the reflection diagram of the fall time. Again the signal reflects back and forth between the line termination until it reaches its final DC value. In both the rise and fall

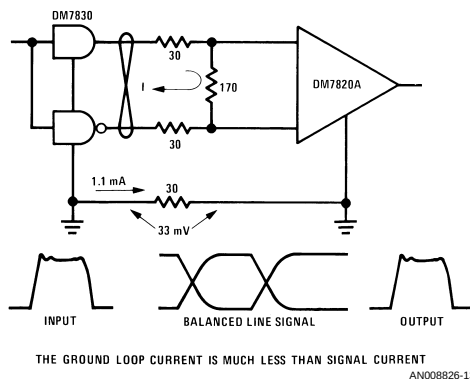


FIGURE 13. Cross Talk of Signals

The circuit used for a line receiver in the balanced method is a differential amplifier. Figure 14 shows a noise transient induced equally on lines A and line B from line C. Because the signals on line A and B are equal, the signals are ignored by the differential line receiver.

Likewise for the same reason, the differential signals on lines A and B from the driver will not induce transients on line C. Thus, the balanced method doesn't generate noise and also isn't susceptible to noise. On the other hand the unbalanced method is more sensitive to noise and also generates more noise.

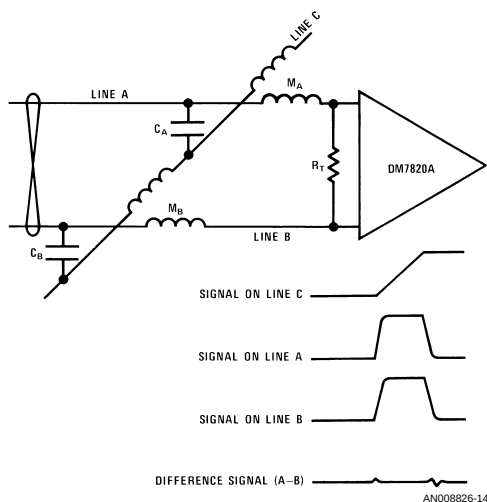


FIGURE 14. Cross Talk of Signals

The characteristic impedance of the unbalanced transmission line is less than the impedance of the balanced transmission line. In the unbalanced method there is more capacitance and less inductance than in the balanced method. In the balanced method the Reactance to adjacent wires is almost cancelled (see Figure 15). As a result a transmission line may have a 60Ω unbalanced impedance and a 90Ω balanced impedance. This means that the unbalanced method, which is more susceptible to IR drop, must use a smaller value termination, which will further increase the IR drop in the line.

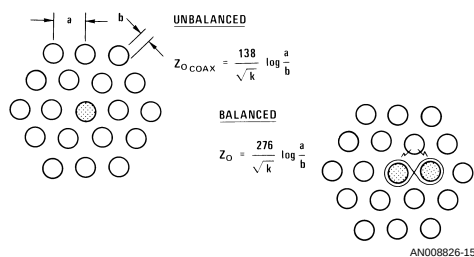


FIGURE 15.  $Z_O$  Unbalanced <  $Z_O$  Balanced

The impedance measurement of an unbalanced and balanced line must be made differently. The balanced impedance must be measured with a balanced signal. If there is any unbalance in the signal on the balanced line, there will be an unbalance reflection at the terminator. Therefore, the lines should also be terminated for unbalanced signals. Figure 16 shows the perfect termination configuration of a balanced transmission line. This termination method is primarily required for accurate impedance measurements.

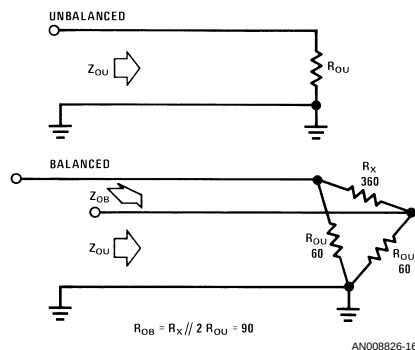


FIGURE 16. Impedance Measurement

#### MEASURED PERFORMANCE

The unbalanced method circuit used in this application note up to this point is the unbalanced circuit shown in Figure 1. The termination of its transmission line was greater than the characteristic impedance of the unbalanced line and the circuit had considerable threshold offset. The measured performance of the unbalanced circuit wasn't comparable to the balanced method. Therefore, for the following comparison of unbalanced and balanced circuits, an improved termination shown in Figure 17 will be used. This circuit terminates the line in 60Ω and minimized the receiver threshold offset.

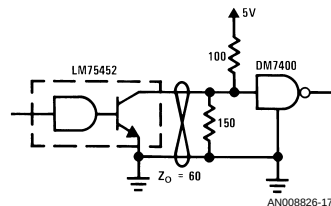


FIGURE 17. Improved Unbalanced Method

A plot of the Absolute Maximum Data Rate versus cable type is shown in Figure 18. The graph shows the different perfor-

mances of the DM7820A line receiver and the DM7830 line driver circuits with a worse case  $\frac{1}{8}$  Duty Cycle in no. 22 AWG stranded wire cables. In a single twisted pair cable there is less reactance than in a cable having nine twisted pairs and in turn this cable has less reactance than shielded pairs. The line length is reduced in proportion to the increased line attenuation which is proportional to the line reactance. The plot shows that the reactance and attenuation has a significant effect on the cable length. Absolute Maximum Data Rate is defined as the Data Rate at which the output of the line receiver is starting to be degraded. The roll off of the performance above 20 mega baud is due to the circuit switching response limitation.

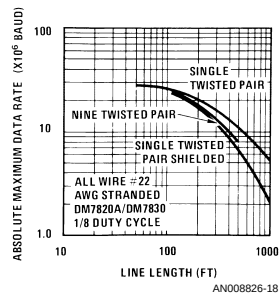


FIGURE 18. Data Rate vs Cable Type

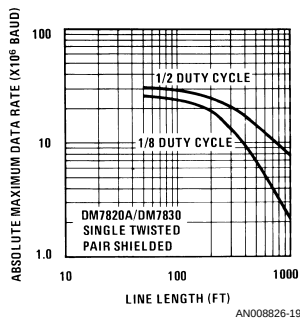


FIGURE 19. Data Rate vs Duty Cycle

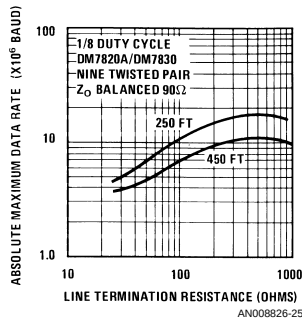


FIGURE 20. Data Rate vs Line Termination

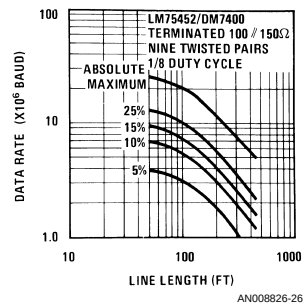


FIGURE 21. Data Rate vs Distortion of LM75452, DM7400

Figure 19 shows the reduction in Data Rate caused by Duty Cycle. It can be observed that the Absolute Maximum Duty Rate of  $\frac{1}{8}$  Duty Cycle is less than  $\frac{1}{2}$  Duty Cycle. The following performance curves will use  $\frac{1}{8}$  Duty Cycle since it is the worst case.

Absolute Maximum Duty Rate versus the Line Termination Resistance for two different lengths of cable is shown in Figure 20. It can be seen from the figure that the termination doesn't have to be perfect in the case of balanced circuits. It is better to have a termination resistor to minimize the extra transient signal reflecting between the ends of the line. The reason the Data Rate increases with increased Termination Resistance is that there is less IR drop in the cable.

The graphs in Figure 21 show the Data Rate versus the Line Length for various percentages of timing distortion using the unbalanced LM75452 and DM7400 circuits shown in Figure 17. The definition of Timing Distortion is the percentage difference in the pulse width of the data sent versus the data received.

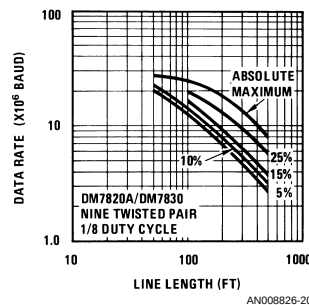


FIGURE 22. Data Rate vs Distortion of DM7820A, DM7830

Data Rate versus the Line Length for various percentage of timing distortion using the balanced DM7820A and DM7830 circuit is shown in Figure 22. The distortion of this method is improved over the unbalanced method, as was previously theorized.

The Absolute Maximum Data Rate versus Line Lengths shown in the previous two figures didn't include any induced

signal noise. Figure 23 shows the test configuration of the unbalanced circuits which was used to measure near end cross talk noise. In this configuration there are eight line drivers and one receiver at one end of the cable. The performance of the receiver measured in the presence of the driver noise is shown in Figure 24.

Figure 24 shows the Absolute Maximum Duty Rate of the unbalanced method versus line length and versus the number of line drivers corresponding to the test configuration delineated in Figure 23. In the noise measurement set-up there was a ground return for each signal wire. If there is only one ground return in the cable the performance is worse. The graph shows that the effective line length is drastically reduced as additional Near End Drivers are added. When this performance is compounded by timing distortion the performance is further reduced.

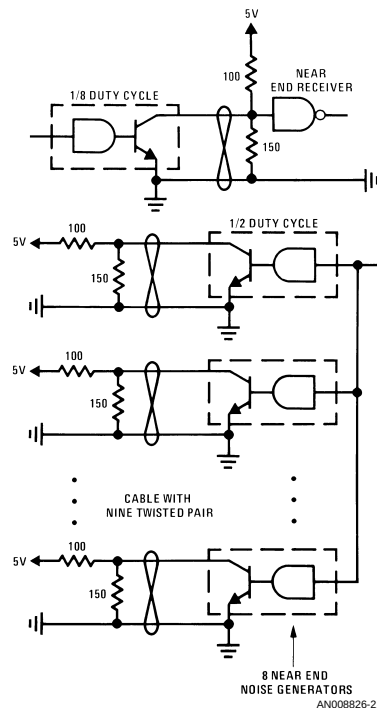


FIGURE 23. Signal Cross Talk Experiment Using DM75452, DM7400

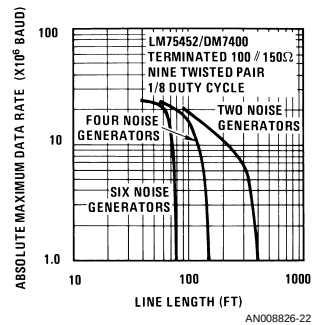


FIGURE 24. Data Rate vs Signal Cross Talk of LM75452, DM7400

Figure 25 shows the test configuration of the balanced circuit used to generate worst case Near End cross talk noise similar to the unbalance performance shown in the previous figure. Unlike the unbalanced case, there was no measurable degradation of the circuits Data Rate or distortion.

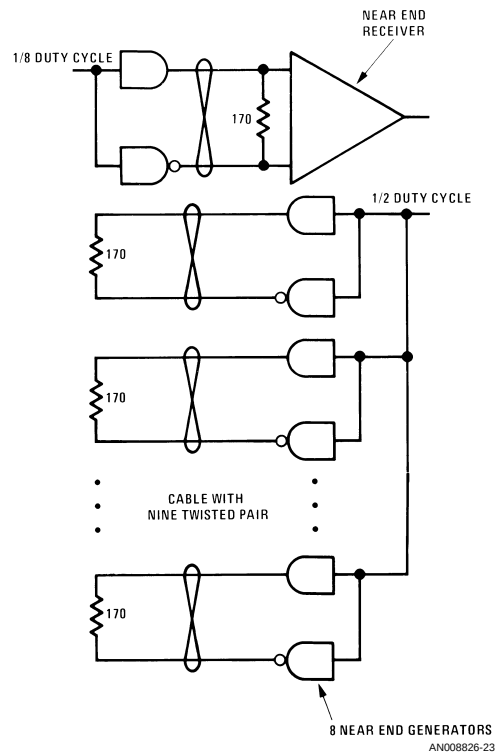


FIGURE 25. Signal Cross Talk Experiment Using DS7830, DS7820A

## CONCLUSION

National has a full line of both Balanced and Unbalanced Line Drivers and Receivers. Both circuit types work well when used within their limitation. This application note shows that the balanced method is preferable for long lines in noisy electrical environments. On the other hand the unbalanced circuit works perfectly well with shorter lines and reduced data rates.

## DEFINITION OF BAUD RATE

Baud Rate  $\equiv$  modulation rate of the channel and is defined as the reciprocal of the minimum pulse width.

Bits/sec (bps)  $\equiv$  information rate of the channel and is defined as the number of bits transmitted in one second.

**Note:** For Non-Return to Zero (NRZ) coding, the baud rate is equal to the bit rate. For Manchester coding, the baud rate is equal to twice the bit rate.

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